

Background

Core theoretical claim – Based on embodied theories: word meaning is grounded in perceptual and motor experiences. Understanding *kick* activates motor cortex | *bitter* activates taste cortex [1,2].

Gap – Most evidence coming from single-word paradigms with isolated stimuli, or relying on LLM-based embeddings, which makes the interpretation difficult [3].

Method – Creating stories based on interpretable Lancaster Sensorimotor Norms (LSN, [4]), which provide a principled 11-dimensional quantification of embodiment across ~40,000 English words.

Intracranial EEG to get high spatial and temporal resolution, focusing on Broadband High Activity (BHA).

Research Question

RQ1: Do different sensorimotor norms recruit spatially distinct cortical regions?

RQ2: At which latencies relative to word onset does sensorimotor content modulate neural activity and do different dimensions show different temporal profiles?

RQ3: Does the sensorimotor signal operate at 3 timescales (word, sentence and story level) consistent with a hierarchical temporal receptive window structure?

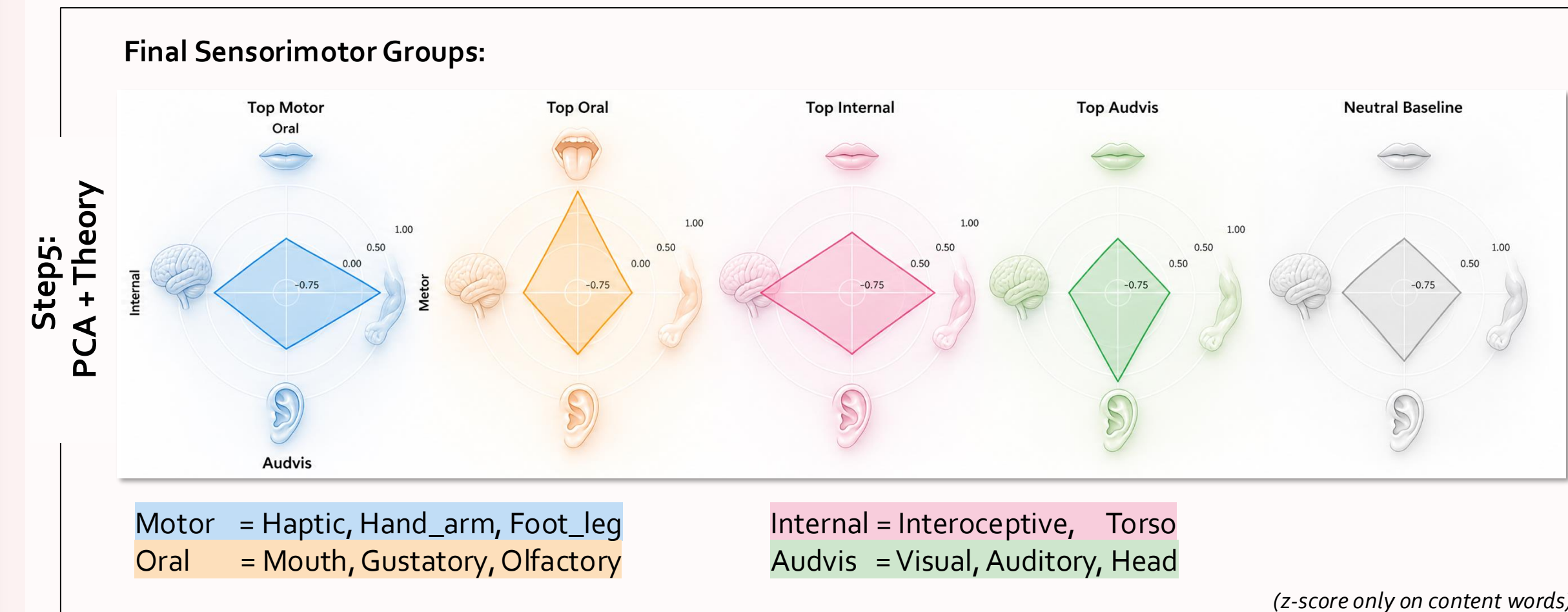
Stimuli Design

11 LSN : The original dimensions on Lancaster Sensorimotor Norms: *Haptic, Hand/arm, Foot_leg, Torso, Mouth, Interceptive, Gustatory, Olfactory, Auditory, Visual, Head;*

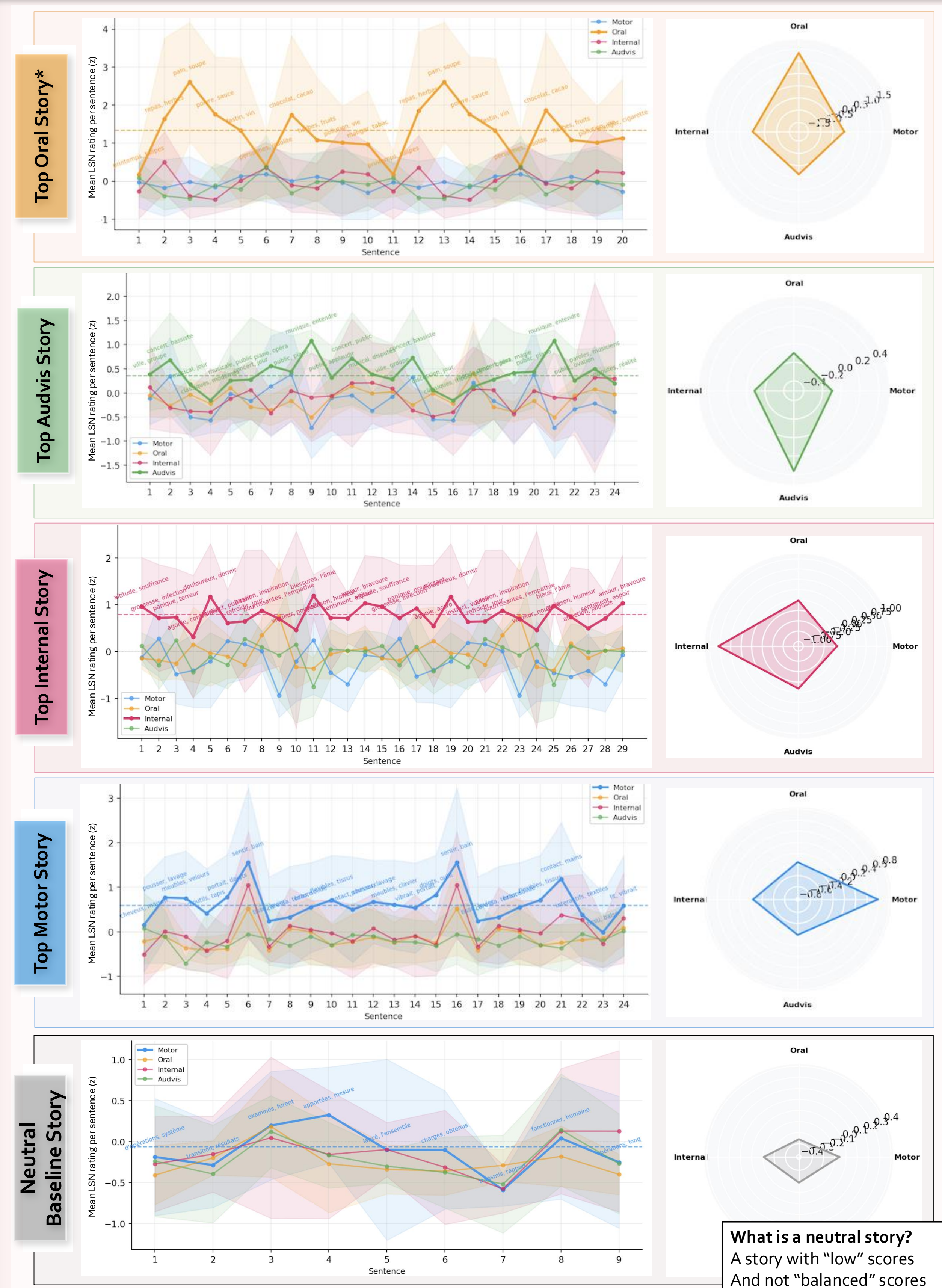
Texts: AI-generated stories based on LSN.

Computation: Story-level score = average word-level LSN across all content in a story.

Steps:	Purist	Subset: Auditory, Haptic, Visual, Motion.	Steps:	No Visual	Auditory, Haptic, Interceptive, Gust+olf, Motion.
		Problem: Visual dominated → Eliminate it			Problem: Motion dominated → Theory-driven PCA + using <i>weighted mean</i>
Steps:	Theory	Audit. = Auditory, Head, Mouth. Gust_olf = Gustatory, Olfactory, Mouth. Haptic = Haptic, Hand_arm. Inter. = Interceptive, Torso. Motion = Foot_leg, Hand_arm, Torso.	Steps:	PCA	PC1 = Haptic, Hand/arm, Visual, Foot_leg, Torso. PC2 = Mouth, Interceptive, Gustatory, Olfactory. PC3 = Gustatory, Olfactory, Foot_leg, Torso. PC4 = Auditory, Visual, Head.



Stimuli Design: 4 Stories per Group + 4 Neutral Stories



Stimuli Validation: Norming Study Structure

Participants: 10 native French speakers

Task: Rating (1–5 Likert scale) **14 short stories + 10 additional sentences** based on 4 sensorimotor groups (+ unnaturalness)

Story pool: 8 selected (2 per group) | 2 neutral: *r.d.f.* 4 stories* | 4 unselected: *r.d.f.* 12 unselected stories*
Sent. pool: Drawn from selected stories the participant did not see during the story rating phase

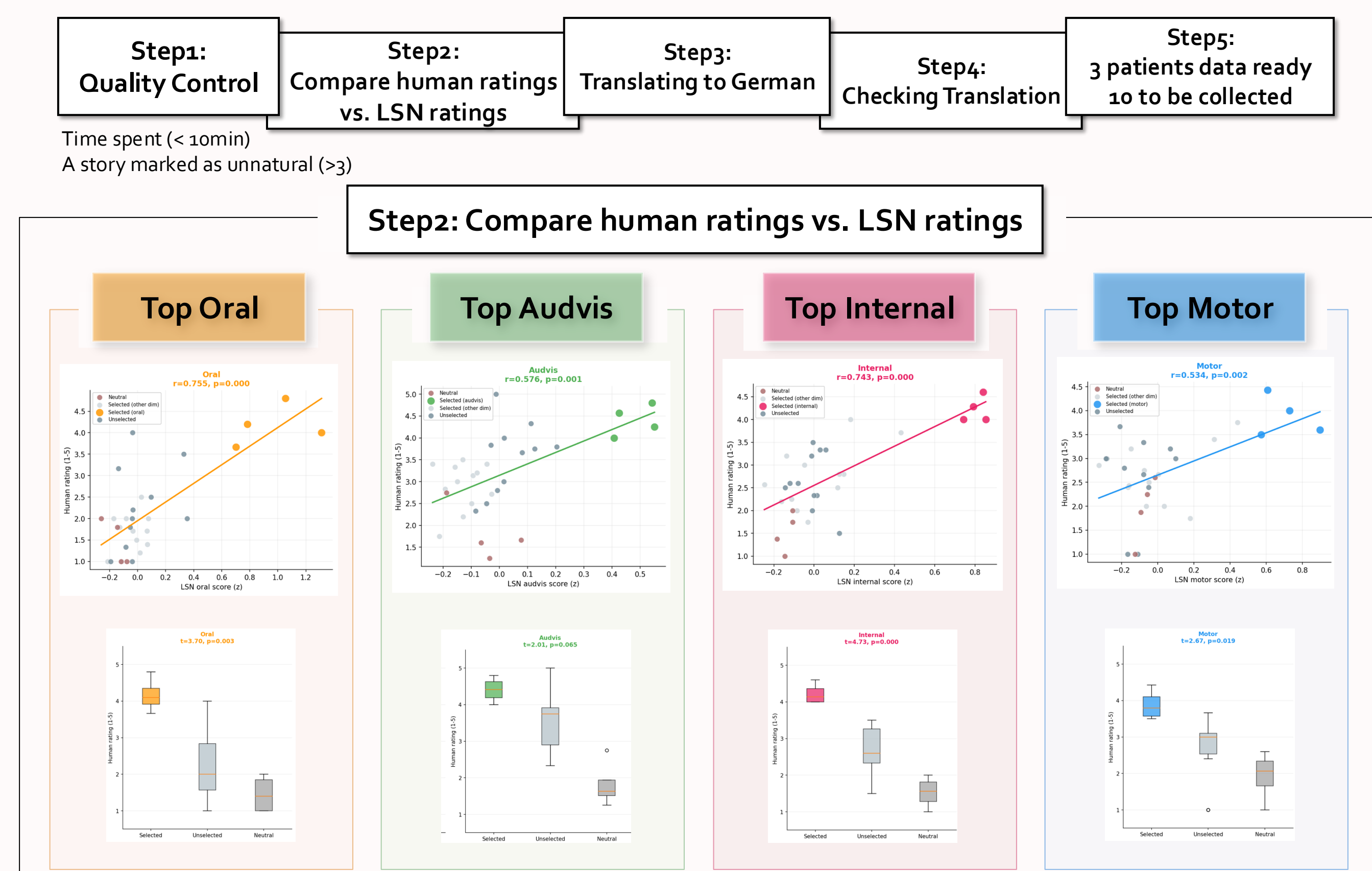
**r.d.f.*: Randomly Drawn From

Selected References:

- [1] Barsalou 1999
[2] Pulvermüller 2013
[3] Huth et al. 2016
[4] Lynott et al. 2020
[5] Martin et al. 2020
[6] Brysbaert et al. 2014
[7] Bonin et al. 2018
[8] Syssau et al. 2021

[Illustrat.co was used to edit the final composition plot for styling purposes only]

Stimuli Validation: Norming Analysis



Planned Analyses

LEVEL 0	PREPROCESSING (used by all 3 levels)	Data Acquisition	Data Alignment	Neural Data Preparation (BHA, 512 HZ, z-scored per subband)	
LEVEL 1	WORD-LEVEL mTRF	Word-level stimulus matrix (n_timestamps, n_features*)	Fit mTRF Lag: -200 to +1000 (?) Ridge regression per electrode	Evaluation R ² per electrode Variance partitioning	Visualizations
LEVEL 2	SENTENCE-LEVEL mTRF	Sentence-level stimulus matrix	Fit mTRF Lag: -500 to +2000 (?) Ridge regression per electrode	Evaluation R ² per electrode Variance partitioning	Visualizations
LEVEL 3	STORY-LEVEL ANALYSIS	Sentence-level feature matrix	Story-level neural responses	Evaluation Ridge regression OR ANOVA Variance partitioning	Visualizations

***Features:** Final sensorimotor groups | Surprisal | Semantic Distance | Concreteness | Valence

Open Questions

FEATURE LEVEL	LEVEL 1: WORD-LEVEL	LEVEL 2: SENTENCE-LEVEL	LEVEL 3: STORY-LEVEL
Semantic Distance	Cosine to context : GPT-2 / FastText ? Window size 3 / 5 / 7 ?	CamemBERT [5] CLS cosine	CamemBERT CLS cosine to prior story?
Surprisal	GPT-2 vs. trigram	Mean/max of content words or language model probability?	
Concreteness	English [6] (40k words), French [7] (~2k)	Mean/max over content words	Mean/max over sentences
Valence	~1k French (FANCat, [8])	Mean/max or CamemBERT CLS?	